

SIPING, China

WMO: 541570

Lat: 43.1203N

Lon: 124.3854E

Elev: 167

StdP: 99.33

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.7	-29.4	0.3	-20.4	-27.3	0.3	-19.3	8.7	-2.6	7.3	-4.4	1.1	200	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	31.2	21.7	30.0	21.5	28.9	21.2	25.0	28.4	24.2	27.5	23.4	26.7	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.0	19.3	27.1	23.2	18.3	26.4	22.4	17.4	25.7	77.5	28.6	74.2	27.7	71.1	26.7	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	7.6	6.4	-26.4	33.3	3.2	1.6	-28.7	34.5	-30.6	35.4	-32.3	36.4	-34.7	37.5		
			-26.5	25.9	3.0	1.2	-28.7	26.8	-30.4	27.4	-32.1	28.1	-34.3	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-12.9	-8.2	-0.1	9.4	16.7	21.9	24.2	22.8	17.1	8.7	-1.6	-10.5	(6)
(7)		DBStd	13.80	4.88	5.68	5.52	5.07	3.99	3.03	2.17	2.63	3.49	4.89	6.19	5.34	(7)
(8)		HDD10.0	2679	711	511	315	71	2	0	0	1	83	349	636		(8)
(9)		HDD18.3	4496	969	744	571	271	79	6	0	3	63	299	597	894	(9)
(10)		CDD10.0	1718	0	0	3	53	211	357	440	396	214	43	2	0	(10)
(11)		CDD18.3	494	0	0	0	2	30	113	182	141	26	1	0	0	(11)
(12)		CDH23.3	3781	0	0	0	22	273	912	1412	980	176	6	0	0	(12)
(13)		CDH26.7	964	0	0	0	2	62	274	383	224	19	1	0	0	(13)
(14)	Wind	WSAvg	2.7	2.2	2.6	3.2	3.7	3.4	2.8	2.5	2.2	2.2	2.6	2.7	2.3	(14)
(15)	Precipitation	PrecAvg	602	5	6	15	29	57	95	149	146	43	29	18	9	(15)
(16)		PrecMax	873	21	21	44	67	200	204	294	331	137	71	58	25	(16)
(17)		PrecMin	380	0	0	0	3	7	20	29	33	6	7	0	0	(17)
(18)		PrecStd	132	4	6	11	18	36	42	64	75	28	18	13	7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	8.4	17.2	26.1	30.6	33.2	32.7	31.8	28.6	23.8	15.9	6.1	(19)
(20)		MCWB	-0.8	2.7	8.6	13.3	16.7	19.8	24.0	23.5	20.6	15.5	10.3	3.6		(20)
(21)		2%	DB	-0.6	5.3	13.3	22.8	28.4	31.2	31.1	30.1	26.9	20.8	12.9	2.4	(21)
(22)		MCWB	-3.0	0.9	6.2	11.9	15.6	19.0	23.6	23.3	18.6	13.7	8.3	0.1		(22)
(23)		5%	DB	-2.8	3.2	11.0	20.6	26.5	29.6	30.0	29.0	25.4	18.7	10.2	0.4	(23)
(24)		MCWB	-4.8	-0.4	4.6	11.2	14.9	18.8	22.9	22.6	17.8	12.4	6.4	-1.5		(24)
(25)		10%	DB	-4.7	1.0	8.6	18.4	24.3	28.1	28.9	27.8	23.9	16.8	7.8	-1.7	(25)
(26)		MCWB	-6.5	-2.2	3.2	9.5	14.0	18.6	22.7	22.0	17.0	11.5	4.8	-3.6		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	3.5	9.5	15.7	19.0	23.0	26.4	26.0	22.6	16.7	10.9	3.8	(27)
(28)		MCDWB	1.2	7.2	15.9	21.5	27.4	28.8	30.5	29.4	26.6	21.3	14.5	5.9		(28)
(29)		2%	WB	-2.9	1.3	7.1	13.3	17.5	21.7	25.2	25.0	20.9	15.1	8.6	0.5	(29)
(30)		MCDWB	-0.9	4.7	12.1	20.9	25.0	27.1	28.8	28.1	24.4	19.3	12.2	2.1		(30)
(31)		5%	WB	-4.8	-0.3	5.1	11.6	16.2	20.9	24.5	24.1	19.6	13.5	6.9	-1.4	(31)
(32)		MCDWB	-2.9	3.0	10.4	19.2	23.3	26.2	28.0	27.1	23.4	17.8	9.8	0.4		(32)
(33)		10%	WB	-6.6	-2.1	3.4	10.2	15.2	20.1	23.8	23.4	18.3	12.2	5.0	-3.7	(33)
(34)		MCDWB	-4.7	1.0	8.3	17.0	22.3	25.2	27.2	26.3	22.3	16.1	7.6	-1.8		(34)
(35)	Mean Daily Temperature Range	MDBR	9.5	9.8	9.6	11.0	10.8	9.3	7.7	8.3	11.2	10.9	8.8	8.9	(35)	
(36)		5% DB	MCDBR	11.5	10.7	12.0	13.4	13.1	11.8	9.3	9.7	12.5	12.7	10.5	10.3	(36)
(37)		MCWBR	9.8	7.7	7.4	6.7	5.4	4.2	3.8	4.2	5.7	6.8	7.2	9.1	(37)	
(38)		5% WB	MCDBR	11.0	10.1	11.2	11.2	10.5	8.8	7.4	7.2	9.6	10.7	9.3	10.3	(38)
(39)		MCWBR	9.6	7.8	7.3	6.5	5.4	3.9	3.8	3.9	5.3	6.7	7.1	9.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.338	0.383	0.455	0.553	0.620	0.689	0.660	0.577	0.497	0.512	0.418	0.355	(40)	
(41)		taud	2.064	1.971	1.862	1.707	1.625	1.575	1.670	1.832	1.946	1.806	1.958	2.031	(41)	
(42)		Ebn at Noon	753	779	767	724	688	643	657	698	723	628	642	683	(42)	
(43)		Edh at Noon	107	140	179	226	252	266	239	197	163	165	120	102	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.30	3.25	4.35	5.13	5.73	5.41	5.02	4.83	4.44	3.22	2.19	1.87	(44)	
(45)		RadStd	0.17	0.23	0.21	0.35	0.44	0.49	0.44	0.38	0.29	0.22	0.20	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page